

Course Assignments Overview

Assignment 1: Image Fundamentals (Unit 1)

- Focus: Image loading, downsampling, quantization (gray levels), RGB channel extraction/correlation, and geometric transforms (rotate, scale, translate).
- Output: Stats, images, and correlation plots.

Assignment 2: Gray-Level & Histograms (Unit 2)

- Focus: Intensity transforms (linear stretching, gamma), global vs. local histogram equalization (CLAHE), and histogram matching.
- Output: Original vs. transformed histograms and images.

Assignment 3: Spatial Filtering (Unit 2)

- Focus: Smoothing (average, Gaussian, median filters) and sharpening (Laplacian, unsharp masking). Analyze noise (Gaussian, salt-and-pepper) using PSNR.
- Output: Visual comparisons and PSNR metrics.

Assignment 4: Frequency-Domain Filtering (Unit 2)

- Focus: Computing 2D DFT (magnitude/phase), implementing low/high-pass filters (Ideal, Gaussian), and comparing frequency-domain Gaussian blur with spatial Gaussian blur.
- Output: Frequency plots and filtered images.

Assignment 5: Edge Enhancement & Kernels (Unit 2)

- *Focus: Analyzing the frequency response of spatial kernels (box, Laplacian, Sobel), designing band-pass filters, and high-frequency emphasis filtering.*
- *Output: Frequency responses and edge-enhanced images.*

Evaluation Grading Rubric

- *70% Practical Implementation: Code correctness, clarity, and plots.*
 - *Breakdown: Unit 1 (20%), Histograms (20%), Filtering & FFT (30%).*
- *30% Written Report: Interpretation of results, parameter selection, method comparisons, and computational complexity.*

Suggestion/Tips:

- *Libraries: Use `skimage.data` for samples or `cv2.imread` for custom images.*
- *Data Types: Pay close attention to conversions between float $[0,1]$ and `uint8`.*

- *Performance:* Crop or downsample large images to speed up processing.
- *Exploration:* Actively vary parameters (cutoffs, kernel sizes, sigma) and document the effects.
- *Submission:* Host and share your implementation code on GitHub.

Lab Assignment Manual: Digital Image Processing

General Instructions:

- *Libraries:* You may use `skimage.data` for standard sample images or `cv2.imread` to load your own custom images.
- *Data Types:* Ensure correct data type conversions between float (range $[0,1]$) and `uint8` (range $[0,255]$) during processing.
- *Performance:* For large images, visualize crops or downsample them to improve processing speed.
- *Exploration:* You are highly encouraged to experiment with different parameters (e.g., cutoffs, sigma values, kernel sizes) and document their effects in your report.
- *Submission:* Submit your written report and provide a link to your implementation code hosted on GitHub.

Assessment & Grading (100% Total):

- **Practical Implementation (70%):** Code correctness, clarity, quality of plots, and in-code explanations. (*Unit 1: 20%, Histograms: 20%, Filtering & FFT: 30%*)
- **Report & Discussion (30%):** A brief report interpreting your visual results, justifying parameter selections, comparing methods, and noting computational complexity.

Lab 1: Image Fundamentals & Visualization (Unit 1)

Objective: Understand basic image arrays, sampling, quantization, color channels, and geometric transformations.

Tasks:

1. **Image Statistics:** Load a color image. Programmatically extract and print its shape, data type, minimum pixel value, maximum pixel value, and mean pixel value.
2. **Sampling:** Downsample the loaded image by a factor of 2, and then by a factor of 4. Display the original and both downsampled images.
3. **Quantization:** Reduce the original image's gray levels to 4, 8, and 16 levels. Display the quantized images and note the visual differences (e.g., false contouring).
4. **Channel Extraction:** Extract the Red, Green, and Blue (R, G, B) channels from the original color image. Display them as separate grayscale images and compute the cross-correlation between these channels.
5. **Geometric Transforms:** Apply the following spatial transformations to your image and display the results:
 - Rotate by 30°
 - Scale by a factor of 0.6
 - Translate by $(tx, ty) = (50, 30)$

Lab 2: Basic Gray-level Transformations & Histogram Processing (Unit 2)

Objective: Implement point processing techniques, histogram equalization, and localized image enhancement.

Tasks:

1. **Intensity Transformations:** Implement linear contrast stretching and gamma correction on a low-contrast image.
2. **Histogram Equalization:** Perform global histogram equalization on an image. Compare your result visually and statistically with adaptive histogram equalization.
3. **Histogram Matching:** Choose a source image (Image A) and a reference image (Image B). Perform histogram matching so that Image A's histogram mimics Image B's histogram. Display both images and their histograms before and after the match.
4. **Local Enhancement:** Apply local contrast enhancement using adaptive histogram equalization. Perform this on each individual color channel (R, G, B) OR solely on the luminance channel of a color space (e.g., HSV/LAB) and merge it back. Compare the outputs.

Lab 3: Spatial Filtering – Smoothing & Sharpening (Unit 2)

Objective: Apply linear and non-linear spatial filters for noise reduction and detail enhancement.

Tasks:

1. **Smoothing Filters:** Apply the following spatial filters to an image and compare the results:
 - Average filter (3x3 and 7x7 kernels)
 - Gaussian filter (sigma = 1.0 and sigma = 2.0)
 - Median filter (3x3 and 7x7 kernels)
2. **Sharpening Filters:** Implement Laplacian sharpening and Unsharp Masking on a slightly blurred image. Compare the resulting edge enhancements.
3. **Noise Evaluation (PSNR):** Add synthetic Gaussian noise to a clean image, then filter it. Repeat this process using salt-and-pepper noise. Compute and report the Peak Signal-to-Noise Ratio (PSNR) between the original clean image and your filtered outputs to evaluate filter performance.

Lab 4: Fourier Transform & Frequency-Domain Filtering (Unit 2)

Objective: Transform images into the frequency domain and apply filters by manipulating frequency components.

Tasks:

1. **2D DFT:** Compute the 2D Fast Fourier Transform (FFT) of a grayscale image. Shift the zero-frequency component to the center of the spectrum and display the log-magnitude and phase plots.

2. **Frequency Filtering:** Implement the following filters and apply them to your image in the frequency domain, then apply the inverse FFT to view the result:
 - Ideal low-pass filter (choose an appropriate cutoff frequency D_0)
 - Gaussian low-pass filter
 - Gaussian high-pass filter
3. **Domain Comparison:** Apply a Gaussian low-pass filter in the frequency domain. Apply an equivalent Gaussian blur in the spatial domain. Compare the two resulting images visually and calculate the PSNR between them.

Lab 5: Filtering Correspondence & Edge Enhancement (Unit 2)

Objective: Analyze spatial kernels in the frequency domain and design advanced frequency filters.

Tasks:

1. **Kernel Frequency Response:** Take small spatial kernels (e.g., Box blur, Laplacian, Sobel). Pad them to the size of your test image, compute their FFT, and display their 2D magnitude responses to visualize what frequencies they affect.
2. **Band-pass Design:** Design a simple band-pass filter in the frequency domain by subtracting a Gaussian low-pass filter with a smaller cutoff from one with a larger cutoff. Apply it to an image and display the result.

3. *High-Frequency Emphasis: Implement high-frequency emphasis filtering using the formula:*

$$g = af + b[f - \text{lowpass}(f)]$$

(where $a \geq 0$ and $b > a$). Apply this to an image and explain how it improves edge enhancement compared to standard high-pass filtering.